

🎯 Objective

Transform theories, ideas, and research into production-scale geospatial, cloud, and data solutions. Passionate about scalable architectures, machine learning, and automation.

⚙️ Skills

Languages: Python, R, JavaScript, SQL, Bash, $\text{\LaTeX}$	Cloud: AWS, GCP, Azure	Data Eng.: Airflow, Kafka, Prefect, Hadoop
Databases: PostgreSQL, MongoDB, MySQL, Snowflake	GIS: ArcGIS, GeoPandas, PostGIS, GDAL, QGIS	DevOps: Docker, Kubernetes, Ansible, GitHub Actions
Visualization: Shiny, Leaflet, Kibana, Plotly	ML/AI: TensorFlow, Scikit-learn, NLP, AI Deployment	Version Control: Git, GitHub, Git-Lab

👛 Experience

Geospatial Systems Architect	<i>Oak Ridge National Lab</i>	2023–Present
Architected cloud-native geospatial pipelines, enhanced data discovery with LLM+RAG implementations, designed a geoparquet-based data warehouse, and developed real-time parcel monitoring with ML models.		
Senior Software Engineer – Data Engineering	<i>Bold Penguin (Remote)</i>	2022–2023
Migrated ETL workflows to Prefect, created OCR microservices, built and maintained microservices, and automated data ingestion pipelines.		
Data Engineer	<i>Oak Ridge National Lab</i>	2018–2022
Led large-scale raster ETL pipeline development, built ML-based misinformation classification models, and optimized geospatial processing from disparate datasets.		
Bioinformatician	<i>Microbial Insights</i>	2016–2018
Managed projects, developed bioinformatics pipelines with automatic report generation, and provided technical support for environmental DNA analysis.		

📁 Projects

This Is A Casino
A machine-learning-driven platform for predicting and trading stocks using semantic learning and reinforcement learning. **Technologies:** Python, Flask, TensorFlow, Scikit-learn, Docker, PostgreSQL, React.

Where I’ve Been
A web application visualizing user travel history with dynamic map interactions and real-time database updates. **Technologies:** Python, FastAPI, PostgreSQL, Leaflet, Vue.js, Bootstrap, Docker.

🎓 Education

M.S. Plant Sciences – Molecular GeneticsUniversity of Tennessee, 2017
B.S. Plant Sciences – BiotechnologyUniversity of Tennessee, 2014